

SIMATS ENGINEERING

ASSESSMENT TOOL 3

NAME: **Siva Dakshitha K**

REG.NO: **192425395**

COURSE NAME: **CSA1603**

COURSE CODE: **Data Warehousing and Data Mining**

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

<i>Criteria</i>	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

I **Siva Dakshitha K / 192425395** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha K

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formalisms [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

CSA1603 - Data Warehousing and Data Mining

C02 - Assessment Tool (3)

Name: Siva Dakshitha K

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Q1: Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it.

Ans: Data Mining:

Data mining is the process of extracting useful patterns and knowledge from large datasets.

Two basic Data Mining Tasks:

- 1) Classification
- 2) Clustering

KDD Steps:

- 1) Data Selection
- 2) Data preprocessing
- 3) Data Transformation
- 4) Data Mining
- 5) Pattern Evaluation

Role of Data Preprocessing:

Data preprocessing removes missing values, noise, and inconsistencies, improving data quality and making data mining results more accurate.

Q2: What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples.

Ans: Data Cleaning:

Data cleaning is the process of removing errors and inconsistencies from data.

Two Data Quality Issues:

1) Missing Values

2) Noisy data

Methods:

- Missing Values: Replace with mean/median or remove the record

Example: Replace a missing age with the average age.

- Noisy data: Use binning or smoothing techniques.

Example: Replace an outlier with the average of nearby values.

Q3: Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration.

Ans: Data Integration:

Data integration is the process of combining data from multiple sources into a single, unified dataset.

Two Challenges:

1) Data redundancy

2) Data inconsistency

Handling Methods:

- Redundancy: Remove duplicate records or attributes.

- Inconsistency: Standardize data formats and correct conflicting values

Q4: What is covariance analysis? List the two covariance formulas.

Ans: Covariance Analysis:
Covariance analysis is used to measure the relationship between two variables and shows whether they change together

Two formulas:

• Population -

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

• Sample -

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). If the stocks are affected by the same industry trends, will their prices rise or fall together and independent?

Ans: Given n

$$X (\text{Stock A}) = \{2, 3, 5, 4, 6\}$$

$$Y (\text{Stock B}) = \{5, 8, 10, 11, 14\}$$

Step 1: Finding Mean

$$\bar{x} = \frac{2+3+5+4+6}{5} \Rightarrow 4$$

$$\boxed{\bar{x} = 4}$$

$$\bar{y} = \frac{5+8+10+11+14}{5} = 9.6$$

$$\boxed{\bar{y} = 9.6}$$

Step 2: calculating Covariance

X	Y	X - 4	Y - 9.6	(X - 4)(Y - 9.6)
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

$$\sum (x - \bar{x})(y - \bar{y}) = 20$$

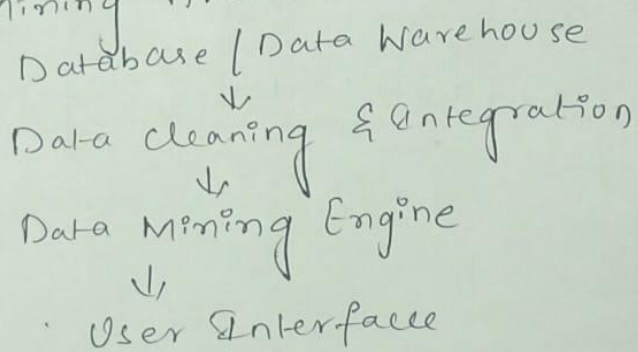
$$\text{Cov}(X, Y) = \frac{20}{5} = 4$$

$$\therefore \text{Cov}(X, Y) = 4$$

$\therefore$ Since Covariance is positive, both stocks rise and fall together. They are not independent.

Q6: Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery.

Ans: Data Mining Architecture:



Components :

- 1) Database / Data Warehouse : Stores large amounts of data.
- 2) Data Cleaning & Integration : Removes errors and combines data from different sources.
- 3) Data Mining Engine : Applies algorithms to find patterns and knowledge.
- 4) Pattern Evaluation : Identifies useful and meaningful patterns.
- 5) User Interface : Displays results to users.

Interaction :

Data is collected, cleaned, processed, mined for patterns, evaluated, and finally presented to users to decision-making.

Q7: Find Covariance for following data set
 $x = \{4, 5, 6, 7, 9\}$ $y = \{8, 3, 7, 5, 6\}$

Ans: Given :

$$x = 4, 5, 6, 7, 9$$

$$y = 8, 3, 7, 5, 6$$

Step 1: Finding mean

$$\bar{x} = \frac{4+5+6+7+9}{5} \Rightarrow \frac{31}{5} \Rightarrow 6.2$$

$$\bar{y} = \frac{8+3+7+5+6}{5} \Rightarrow \frac{29}{5} \Rightarrow 5.8$$

$$\left. \begin{array}{l} \bar{x} = 6.2 \\ \bar{y} = 5.8 \end{array} \right\}$$

Step 2: Creating a table

X	Y	$x - \bar{x}$	$y - \bar{y}$
4	8	-2.2	2.2
5	3	-1.2	-2.8
6	7	-0.2	1.2
7	5	0.8	-0.8
9	6	2.8	0.2

Step 3: Apply the Covariance formula.

• population Covariance

$$\begin{aligned}\text{Cov}(x, y) &= \frac{\sum (x - \bar{x})(y - \bar{y})}{n} \\ &= \frac{-1.80}{5} \Rightarrow -0.36\end{aligned}$$

$$\therefore \text{Cov}(x, y) = -0.36$$

• Sample Covariance

$$\begin{aligned}\text{Cov}(x, y) &= \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1} \\ &= \frac{-1.80}{5-1} \Rightarrow \frac{-1.80}{4} \Rightarrow -0.45\end{aligned}$$

$$\therefore \text{Cov}(x, y) = -0.45$$

∴ The covariance is negative, which means x & y have a weak negative relationship. As one variable increases the other tends to decrease slightly.

Q.8: Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and preprocessing inconsistent formats, propose a systematic pre processing pipeline and justify the sequence of steps.

Ans: Major Steps in Data Pre-processing:

1) Data Cleaning - Handle missing values, noise, and outliers.

2) Data Integration - Combine data from different sources.

3) Data Transformation - Normalize and convert data into suitable formats.

4) Data Reduction - Reduce data size while maintaining important information.

Pre-processing pipeline:

- 1) Remove inconsistencies and handle missing values.
- 2) Detect and remove outliers.
- 3) Standardize data formats.
- 4) Transform and normalize data.
- 5) Reduce unnecessary data.

Justification:

The Sequence improves data quality step-by-step and ensures accurate and reliable data mining results.

Q.9: A hospital is analyzing patient age data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using equal-frequency (equi-depth) binning.

Ans: Given data:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Total no. of values = 27

Using equal-frequency binning, divide data into 3 bins (9 values each)

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18.0

Bin 2: 22, 25, 25, 25, 25, 30, 33, 35

Mean = 28.11

Smoothed Bin 2:

28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11

Bin 3:

35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 43.77

Smoothed Bin 3:

43.77, 43.77, 43.77, 43.77, 43.77, 43.77, 43.77, 43.77, 43.77

Conclusion:

Equal-frequency binning smooths the data by replacing values with their bin mean, reduces noise.

Q.10: Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples.

Ans: Data Discretization:

It is the process of converting continuous data into groups or intervals.

Importances:

- Reduces data complexity.
- Helps in better data analysis.

Types:

- Equal-Width Binning-

Divides data into equal-sized intervals

Example: 0-10, 10-20, 20-30

- Equal-Frequency Binning-

Divides data into equal number of values in each group.

Impact:

Equal width is simple, while equal-frequency gives balanced groups and better results.